AP PRECALCULUS

Unit 1 • Worksheet 1.1-1.6C • Answer key

# Review 1.1-1.6C • Review of Topics 1.1 to 1.6

Answer key with worked solutions

**Directions**

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

### Question 1 (1.2)

NO CALCULATOR

$P(t)$ is the number of bacteria after $t$ hours, $P(2) = 500$, $P(5) = 1400$. What is the average rate of change of $P$ on $[2, 5]$?

(A) 300 bacteria per hour

(B) 900 bacteria per hour

(C) 280 bacteria per hour

(D) 180 bacteria per hour

**Solution**

**Answer: (A).** $\frac{900}{3} = 300$.

### Question 2 (1.4)

NO CALCULATOR

A polynomial is increasing on $(-\infty, -1)$, decreasing on $(-1, 4)$ and increasing on $(4, \infty)$. Which is true?

(A) local maximum at $x = 4$

(B) local minimum at $x = -1$

(C) local maximum at $x = -1$

(D) no local extrema

**Solution**

**Answer: (C).** Increasing to decreasing at $x = -1$.

### Question 3 (1.5)

NO CALCULATOR

What is the multiplicity of the zero $x = -2$ of $p(x) = (x^2 - 4)^2(x + 2)$?

(A) 1

(B) 2

(C) 3

(D) 5

**Solution**

**Answer: (C).** $(x^2 - 4)^2 = (x - 2)^2(x + 2)^2$, times $(x + 2)$: $(x + 2)^3$.

**Question 4** (1.6) NO CALCULATOR

Which describes the end behavior of $p(x) = -x^2(x-1)(x+5)^2$?

(A) up on both ends
(B) down on both ends
(C) up on the left, down on the right
(D) down on the left, up on the right

**Solution**
**Answer: (C).** Leading term $-x^5$.

**Question 5** (Free response) CALCULATOR

The volume of a box made from a 20 cm by 30 cm sheet by cutting squares of side $x$ cm is $V(x) = x(30-2x)(20-2x)$.

(a) What is the domain of $V$ in this context? Explain.
(b) Find $V(3)$ and interpret it.
(c) Use a graphing calculator to find the value of $x$ that maximizes $V$ and the maximum volume.
(d) Find the average rate of change of $V$ on $[1,3]$, with units.

**Solution**
(a) $0 < x < 10$: the side lengths $x$, $30-2x$, $20-2x$ must be positive.
(b) $V(3) = 3 \cdot 24 \cdot 14 = 1008$: squares of side 3 cm give a box of 1008 $\text{cm}^3$.
(c) $x \approx 3.924$ cm; $V \approx 1056.306$ $\text{cm}^3$.
(d) $V(1) = 1 \cdot 28 \cdot 18 = 504$: $\frac{1008-504}{2} = 252$ $\text{cm}^3$ per cm.